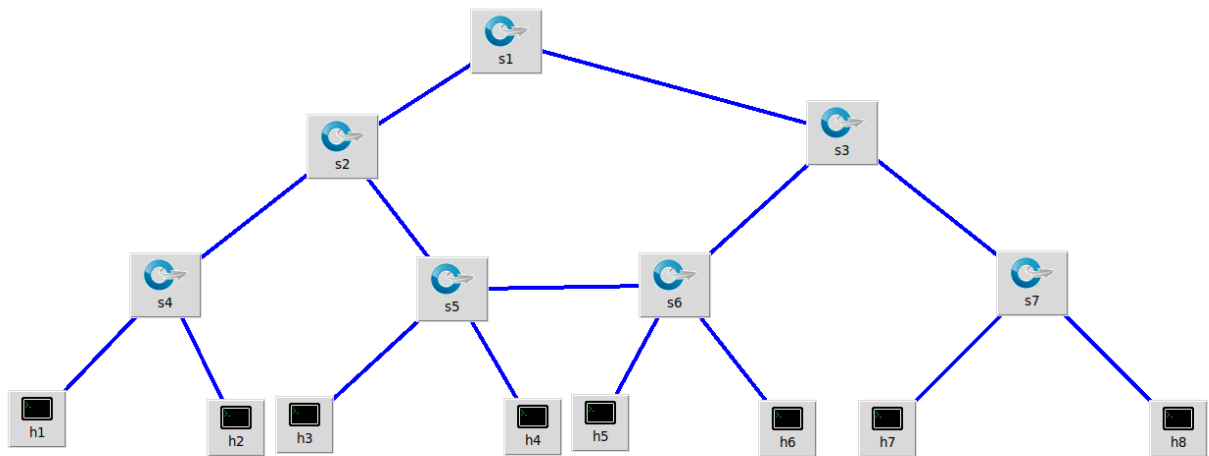


National University Laguna
Midterm Exam, Term 2, AY 2025-2026
SWDENET – Software Defined Networks and Virtualization

Instructions:

1. Do not modify the format of this document.
2. Put your answers in the spaces provided in each item.
3. Upload this document with the filename format:
MidtermExam_SWDENET_Lastname_Firstname.pdf

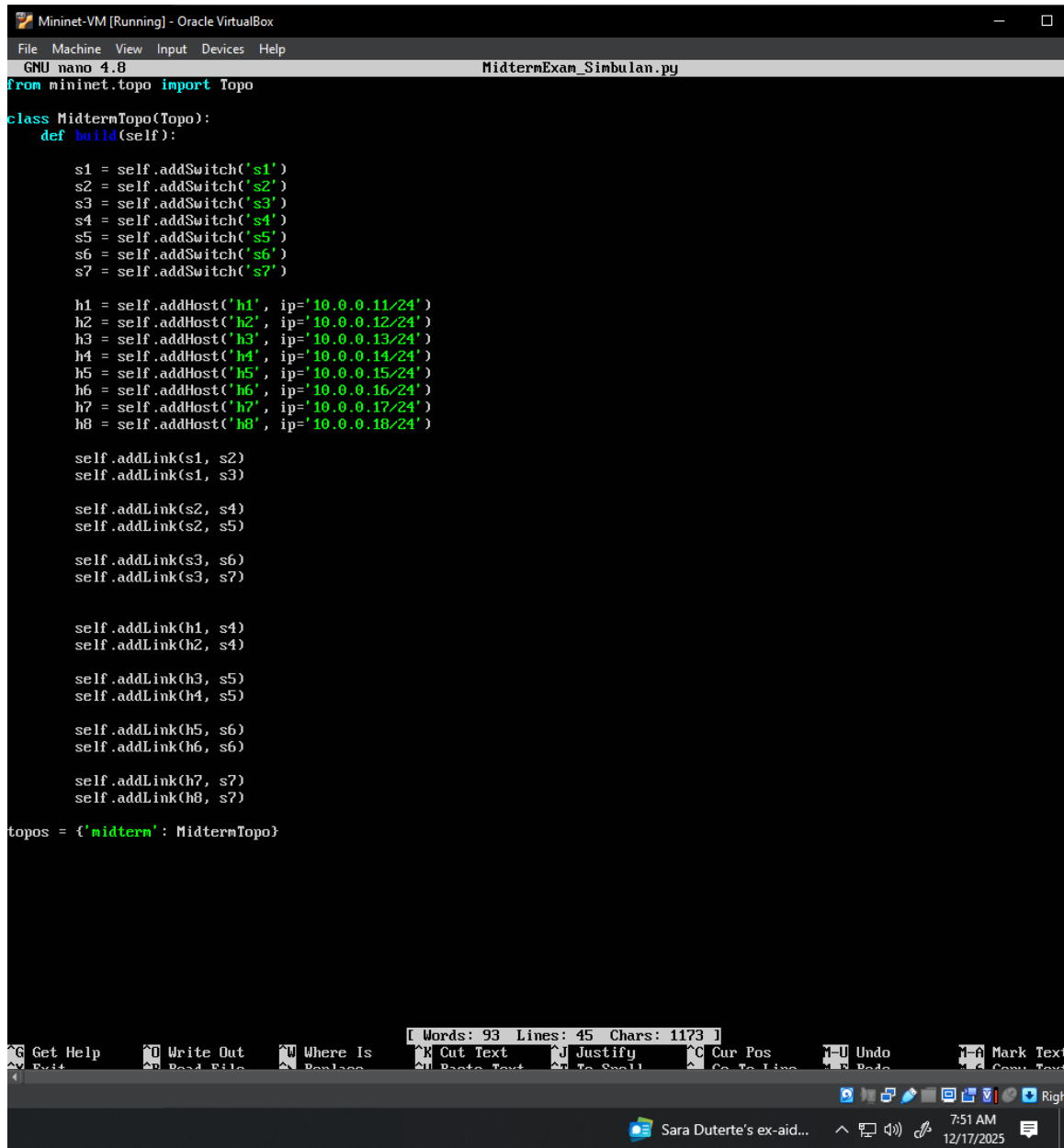
Tasks:



1. Create a custom topology that satisfies the network diagram above.
 - `s1...s7` are switches
 - `h1...h8` are hosts
 - only `s4`, `s5`, `s6`, and `s7` have hosts connected on them.
2. Set the IP address of `h1...h8` following the sequence `10.0.0.11 ... 10.0.0.18`

Requirements:

1. Paste a screenshot of your python code in a NANO environment. Make sure the date and time of your computer is captured.



```
Mininet-VM [Running] - Oracle VirtualBox
File Machine View Input Devices Help
GNU nano 4.8 MidtermExam_Simbulan.py
from mininet.topo import Topo

class MidtermTopo(Topo):
    def build(self):

        s1 = self.addSwitch('s1')
        s2 = self.addSwitch('s2')
        s3 = self.addSwitch('s3')
        s4 = self.addSwitch('s4')
        s5 = self.addSwitch('s5')
        s6 = self.addSwitch('s6')
        s7 = self.addSwitch('s7')

        h1 = self.addHost('h1', ip='10.0.0.11/24')
        h2 = self.addHost('h2', ip='10.0.0.12/24')
        h3 = self.addHost('h3', ip='10.0.0.13/24')
        h4 = self.addHost('h4', ip='10.0.0.14/24')
        h5 = self.addHost('h5', ip='10.0.0.15/24')
        h6 = self.addHost('h6', ip='10.0.0.16/24')
        h7 = self.addHost('h7', ip='10.0.0.17/24')
        h8 = self.addHost('h8', ip='10.0.0.18/24')

        self.addLink(s1, s2)
        self.addLink(s1, s3)

        self.addLink(s2, s4)
        self.addLink(s2, s5)

        self.addLink(s3, s6)
        self.addLink(s3, s7)

        self.addLink(h1, s4)
        self.addLink(h2, s4)

        self.addLink(h3, s5)
        self.addLink(h4, s5)

        self.addLink(h5, s6)
        self.addLink(h6, s6)

        self.addLink(h7, s7)
        self.addLink(h8, s7)

topos = {'midterm': MidtermTopo}
```

Words: 93 Lines: 45 Chars: 1173

Get Help Write Out Where Is Cut Text Justify Cur Pos Undo Mark Text

Sara Duterte's ex-aid... 7:51 AM 12/17/2025

2. Run a `pingall` command. All hosts should be able reach one another. Take a screenshot and paste it below.

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 h4 h5 h6 h7 h8
h2 -> h1 h3 h4 h5 h6 h7 h8
h3 -> h1 h2 h4 h5 h6 h7 h8
h4 -> h1 h2 h3 h5 h6 h7 h8
h5 -> h1 h2 h3 h4 h6 h7 h8
h6 -> h1 h2 h3 h4 h5 h7 h8
h7 -> h1 h2 h3 h4 h5 h6 h8
h8 -> h1 h2 h3 h4 h5 h6 h7
*** Results: 0% dropped (56/56 received)
mininet> _
```

3. Run a dump command, take a screenshot, and paste it below.

```
mininet> dump
<Host h1: h1-eth0:10.0.0.11 pid=9869>
<Host h2: h2-eth0:10.0.0.12 pid=9871>
<Host h3: h3-eth0:10.0.0.13 pid=9873>
<Host h4: h4-eth0:10.0.0.14 pid=9875>
<Host h5: h5-eth0:10.0.0.15 pid=9877>
<Host h6: h6-eth0:10.0.0.16 pid=9879>
<Host h7: h7-eth0:10.0.0.17 pid=9881>
<Host h8: h8-eth0:10.0.0.18 pid=9883>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=9888>
<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None,s2-eth3:None pid=9891>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None,s3-eth3:None pid=9894>
<OVSSwitch s4: lo:127.0.0.1,s4-eth1:None,s4-eth2:None,s4-eth3:None pid=9897>
<OVSSwitch s5: lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None pid=9900>
<OVSSwitch s6: lo:127.0.0.1,s6-eth1:None,s6-eth2:None,s6-eth3:None pid=9903>
<OVSSwitch s7: lo:127.0.0.1,s7-eth1:None,s7-eth2:None,s7-eth3:None pid=9906>
<Controller c0: 127.0.0.1:6653 pid=9862>
mininet>
```